

Your Paper

You

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Abstract

Your abstract.

1 Introduction

Some relevant references:

- VAR-models: [Nicholson et al. \(2020\)](#), including some theory that builds on [Basu and Michailidis \(2015\)](#).
- VARX-models: [Nicholson et al. \(2017\)](#); [Wilms et al. \(2017\)](#)
- VARMA-models: [Wilms et al. \(2023\)](#); [Düker et al. \(2024\)](#)

For $k, m \in \mathbb{N}$ and sample size $T \in \mathbb{N}$, we observe two time series $(\mathbf{y}_t)_{t \in [T]}$ with $\mathbf{y}_t = (y_{t,1}, \dots, y_{t,k})^\top \in \mathbb{R}^k$ and $(\mathbf{x}_t)_{t \in [T]}$ with $\mathbf{x}_t = (x_{t,1}, \dots, x_{t,m})^\top \in \mathbb{R}^m$. The two time series are assumed to follow, for some unknown $(p, q) \in \mathbb{N}_0^2$, a *vector-autoregression with exogenous variables of order (p, q)* , notation $\text{VARX}(p, q)$, that is, there exists $\boldsymbol{\mu} \in \mathbb{R}^k$, $\Phi^{(1)}, \dots, \Phi^{(p)} \in \mathbb{R}^{k \times k}$, $B^{(1)}, \dots, B^{(q)} \in \mathbb{R}^{k \times m}$ and a non-singular covariance matrix $\Sigma_\varepsilon \in [0, \infty)^{k \times k}$ such that

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{\ell \in [p]} \Phi^{(\ell)} \mathbf{y}_{t-\ell} + \sum_{j \in [q]} B^{(j)} \mathbf{x}_{t-j} + \varepsilon_t \quad (1)$$

for some k -dimensional white-noise process $(\varepsilon_t)_t$ with $\varepsilon_t = (\varepsilon_{t,1}, \dots, \varepsilon_{t,k})^\top$ and $\text{Cov}(\varepsilon_t) = \Sigma_\varepsilon$. We write $\Phi = (\Phi^{(1)}, \dots, \Phi^{(p)})$ and $B = (B^{(1)}, \dots, B^{(q)})$.

In a spatial application, the coordinates of $\mathbf{y}$ correspond to spatial locations. In that case, we denote the spatial locations by $s_1, \dots, s_k$ and write $\mathbf{y}_t = (y_t(s_1), \dots, y_t(s_k))^\top$. Some coordinates of the exogenous variables may correspond to spatial locations as well: we write $\mathbf{x}_t(s_j) = (x_{t,1}(s_j), \dots, x_{t,m_s}(s_j))^\top$ for the $m_s \in \mathbb{N}_0$ exogenous variables at location s_j , and $\mathbf{g}_t \in \mathbb{R}^{m_g}$ for potential additional (global) exogenous variables that do not depend on the spatial location, where $m_g \in \mathbb{N}_0$. Overall, the vector $\mathbf{x}_t$ is hence given by

$$\mathbf{x}_t = (\mathbf{x}_t(s_1)^\top, \dots, \mathbf{x}_t(s_k)^\top, \mathbf{g}_t^\top)^\top \in \mathbb{R}^{k m_s + m_g}.$$

Depending on the specific application and inspired by standard SAR-modeling ([Cressie, 1993](#), Section 6), it may in fact be useful to include a summand for $\ell = 0$ in (1), with the respective parameter matrix $\Phi^{(0)}$ being zero on the diagonal; and likewise for $j = 0$. We may then face similar identifiability problems as in structural VAR models, see [Beenstock and Felsenstein \(2007\)](#).

[AB: In econometrics, VAR-models are often employed for residual dependencies. For the model above, this would mean to instead impose that

$$\left(\mathbf{y}_t - \boldsymbol{\mu} - \sum_{j \in [q]} B^{(j)} \mathbf{x}_{t-j} \right) = \sum_{\ell \in [p]} \Phi^{(\ell)} \left(\mathbf{y}_{t-\ell} - \boldsymbol{\mu} - \sum_{j \in [q]} B^{(j)} \mathbf{x}_{t-\ell-j} \right) + \varepsilon_t,$$

which seems to be a slightly more restrictive version of the model in (1).]

Assuming the $\text{VARX}(p, q)$ -model, we are interested in estimating the parameters $(\boldsymbol{\mu}, \Phi, B)$, which are, for instance, crucial for prediction; see [Nicholson et al. \(2017\)](#). For that purpose, we may rely on penalized least-squares estimation:

$$(\hat{\boldsymbol{\mu}}, \hat{\Phi}, \hat{B}) \in \operatorname{argmin}_{\boldsymbol{\mu}, \Phi, B} \sum_{t \in [T]} \left\| \mathbf{y}_t - \boldsymbol{\mu} - \sum_{\ell \in [p]} \Phi^{(\ell)} \mathbf{y}_{t-\ell} - \sum_{j \in [q]} B^{(j)} \mathbf{x}_{t-j} \right\|_F^2 + \mathcal{P}_\lambda(\boldsymbol{\mu}, \Phi, B),$$

where $\|A\|_F^2 = \sum_{i,j} A_{ij}^2$ denotes the squared Frobenius norm of matrix (or vector) A and where $\mathcal{P}_\lambda(\boldsymbol{\mu}, \Phi, B) \geq 0$ denotes a penalty function depending on a vector of penalty parameters $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_L)^\top \in [0, \infty)^L$, with $L \in \mathbb{N}$. In a non-spatial setting, [Nicholson et al. \(2017\)](#) consider several structured choices for $\mathcal{P}_\lambda(\boldsymbol{\mu}, \Phi, B) = \lambda[\mathcal{P}_y(\Phi) + \mathcal{P}_x(B)]$, all based on $L = 1$ (see their Table 1). For instance, the sparse lag group penalty is defined by

$$\mathcal{P}_\lambda(\boldsymbol{\mu}, \Phi, B) = \lambda \left[(1 - \alpha) \left(k \sum_{\ell \in [p]} \|\Phi^{(\ell)}\|_F + \sqrt{k} \sum_{j \in [q]} \sum_{i \in [m]} \|B_{\cdot i}^{(j)}\|_F \right) + \alpha (\|\Phi\|_1 + \|B\|_1) \right],$$

where $\alpha \in [0, 1]$ is some additional hyperparameter and where $\|\cdot\|_1$ denotes the sum over all absolute values of entries of a tuple of matrices or vectors. Recall that penalization based on the Frobenius norm, such as $\|\Phi^{(\ell)}\|_F$, forces all entries of $\Phi^{(\ell)}$ to be either simultaneously zero or simultaneously non-zero.

Potential topics/goals for a paper:

- **[AG: Adaptive weights:** extend [Nicholson et al. \(2017\)](#) by adaptive weights to increase flexibility of the penalization; in principal, the idea here is to add to each penalty term an individual weight which is exactly the inverse of the penalty evaluated at an unpenalized maximum likelihood estimate (if it exists; or a slightly ridge-penalized version instead; see, e.g., [Zou and Hastie, 2006](#)). For the classical lasso, for example, the penalty would be $\lambda \sum_{j=1}^P 1/|\hat{\beta}_j^{ML}| \cdot |\beta_j|$]
- Empirical investigation of penalty functions that are designed to tackle spatial applications. Would need to study spatial literature, and connect it to [Nicholson et al. \(2017\)](#) and others. Potentially develop theory as well. If not studied elsewhere yet under a different name, the model could be called STARX (Spatio-temporal autogression with exogenous variables).

An attempt: for some small spatial distance h , let

$$N_h = \{(i, j) : \|s_i - s_j\|_2 \leq h\} \subset [k]^2$$

denote the set of index pairs for which the spatial distance between s_i and s_j is not larger than h . Akin to the motivation for the *own/other group lasso* in [Nicholson et al. \(2017\)](#), $y_t(s_i)$ is more likely to depend on lagged values of $y_{t-1}(s_j), \dots, y_{t-p}(s_j)$ for j such that $(i, j) \in N_h$ than for j such that $(i, j) \in N_h^c := [k]^2 \setminus N_h$ (Tobler's first law of geography). In the case of no exogenous variables, this would give rise to a spatial group lasso of the form

$$\mathcal{P}_\lambda(\Phi) = \lambda \left[\sqrt{|N_h|} \sum_{\ell \in [p]} \|\Phi_{N_h}^{(\ell)}\|_F + \sqrt{k^2 - |N_h|} \sum_{\ell \in [p]} \|\Phi_{N_h^c}^{(\ell)}\|_F \right],$$

where $\Phi_N^{(\ell)}$ is the stacked vector containing all $\Phi_{i,j}^{(\ell)}$ with $(i, j) \in N$. In the presence of exogenous variables, one may use

$$\begin{aligned} \mathcal{P}_\lambda(\boldsymbol{\mu}, \Phi, B) = & \lambda \left[\sum_{\ell \in [p]} \left\{ \sqrt{|N_h|} \cdot \|\Phi_{N_h}^{(\ell)}\|_F + \sqrt{k^2 - |N_h|} \cdot \|\Phi_{N_h^c}^{(\ell)}\|_F \right\} \right. \\ & \left. + \sum_{j \in [q]} \left\{ \sqrt{|N_0|} \cdot \|B_{N_0}^{(j)}\|_F + \sqrt{|G|} \cdot \|B_G^{(j)}\|_F + \sqrt{mk - |N_0| - |G|} \cdot \|B_{(N_0 \cup G)^c}^{(j)}\|_F \right\} \right] \end{aligned}$$

where $B_{N_0}^{(j)}$ is the stacked vector of all entries of $B^{(j)}$ which correspond to 'the same spatial locations' (i.e., the parameters that link $\mathbf{x}_{t-j}(s_i)$ to $y_t(s_i)$, for some $i \in [k]$; for instance, if $m_s = 1$, $B_{N_0}^{(j)}$ contains all entries $B_{ii}^{(j)}$ with $i \in [k]$), where $B_G^{(j)}$ is the stacked vector containing the last m_g columns of $B^{(j)}$ (the parameters linking the global variables to $\mathbf{y}_t$) and where $B_{(N_0 \cup G)^c}^{(j)}$ are the remaining entries.

- **Design of (spatial) fusion penalties.** Following [Gertheiss and Tutz \(2010\)](#), we can define certain fusion-type penalties to fuse coefficients which exhibit small spatial distances. For example, relying on the same set N_h from above, and again, for the case of no exogenous variables, this would give rise to a spatial fusion lasso of the form

$$\mathcal{P}_\lambda(\Phi) = \lambda \left[\sum_{\ell \in [p]} \left(\sum_{\substack{(j,k) \in N_h; m \in [k] \\ j \neq k \neq m \neq j}} |\Phi_{j,m}^{(\ell)} - \Phi_{k,m}^{(\ell)}| \right) + \left(\sum_{\substack{(j,k) \in N_h; m \in [k] \\ j \neq k \neq m \neq j}} |\Phi_{m,j}^{(\ell)} - \Phi_{m,k}^{(\ell)}| \right) \right], \quad (2)$$

This seems to be a special case of

$$\mathcal{P}_\lambda(\Phi) = \lambda \left[\sum_{\ell \in [p]} \sum_{(j,k,m,n) \in [k]^4} \omega_{j,k,m,n} |\Phi_{j,m}^{(\ell)} - \Phi_{k,n}^{(\ell)}| \right],$$

where $\omega_{j,k,m,n}$ are non-negative weights. Namely,

$$\begin{aligned} w_{j,k,m,n} = & \mathbb{1}((j,k) \in N_h \text{ and } m = n \text{ and } j, k, m \text{ pairwise different}) \\ & + \mathbb{1}((m,n) \in N_h \text{ and } j = k \text{ and } j, m, n \text{ pairwise different}). \end{aligned}$$

The formulation with weights allows for including spatial distances in a smooth way, for instance as in Section 4.2 in [Gertheiss and Tutz \(2010\)](#). fusing (in both spatial dimensions) those coefficients from $\Phi^{(\ell)}$ corresponding to spatial locations s_j and s_k if both belong to the set of small distance N_h . As long as all individual penalty components are regarded separately and shall receive exactly the same “importance”, no grouping weights are needed. However, note that if e.g. for the two dimensions of $\Phi^{(\ell)}$, i.e. its rows and columns, two different neighborhood sets N_h and $N_{\tilde{h}}$ shall be regarded, the two major terms from equation (2) involve different numbers of penalty components and one has to adjust for the respective degrees of freedom by including respective weights, if still a single penalty parameter shall be used. The same applies in case a similar fusion penalty shall be included for exogenous variables. A thorough instruction of how these weights have to be designed can be found in [Bondell and Reich \(2009\)](#). In principle, the use of these weights can be motivated through standardization of the corresponding design matrix part, in analogy to standardization of metric predictors. Furthermore, again adaptive versions of the weights can be used. Other than that, alternative weighting schemes are discussed in the literature; see, for example, [Chiquet et al. \(2017\)](#).

- Adaptive validation for tuning parameter selection?
- Florian Ziel and Co-Authors have presented work where $(\mathbf{y}_t)_t$ is 24-dimensional time series, corresponding to hourly energy prices collected over a day, and explicitly use some from VARX-modelling, with the exogenous variables including wind speed forecasts etc. See for instance [Ziel and Weron \(2018\)](#). The models include some form of non-linearity though. Maybe still relevant as an applied starting point? [AB: Some further related references, who also discuss models of the above form, but under different names: [Beenstock and Felsenstein \(2007\)](#); [Elhorst et al. \(2021\)](#). A keyword seems to be spatial econometrics, see [Elhorst \(2014\)](#), and in particular the model in Section 4.2.]
- [AJ: [Wang et al. \(2023\)](#) asymptotic theory for a spatio-temporal penalized VAR but without exogenous variables, [Hu et al. \(2025\)](#) introduces neighborhood information into VAR estimation via hard nearest-neighbor groups, [Safikhani et al. \(2020\)](#) a generalized STAR model in a high-dimensional empirical application]
- [IW: Additional literature on spatio-temporal autoregressions: [Reuvers and Wijler \(2024\)](#), which was inspired on [Dou et al. \(2016\)](#) and [Gao et al. \(2019\)](#). Other references: [Ma et al. \(2023\)](#), [Pu et al. \(2025\)](#); [Zhu et al. \(2026\)](#) (low rank as alternative dimension reduction to sparsity), [Huang et al. \(2026\)](#) (though they focus on inference).]
- Potential data application: bike rental, Innsbruck weather / lightning data, ...

[AG: **Rental bike data:** A quick ChatGPT search revealed there is several ones which could be useful for us:

1. Capital Bikeshare (Washington D.C.): <https://capitalbikeshare.com/system-data>; weather data sets from NOAA or OpenWeather are often merged to obtain temperature, precipitation, humidity, wind speed; should also be available on Kaggle
2. Citi Bike (New York City): <https://citibikenyc.com/system-data>; very large and detailed dataset with station-level spatial information; weather can be merged from NOAA datasets. [IW: Easy example dataset constructed from Citi Bike available here : https://github.com/MarieTernes/ml-crosstemporal-reconciliation/tree/main/code_INTJOF. This is the data we used in [Rombouts et al. \(2025\)](#).]

3. Divvy Bikes (Chicago): <https://divvy-tripdata.s3.amazonaws.com/index.html>; spatio-temporal demand modeling; weather again can be merged externally
4. [IW: ENTSO-E Transparency Platform for energy data <https://transparency.entsoe.eu/>]

Maybe this helps to perform a more detailed data search, ideally finding those data set in an extended form, where someone has already merged wheather info.]

[AJ: some more literature from Johannes: https://scholar.google.com/citations?view_op=view_citation&hl=fr&user=M8h_FPcAAAAJ&citation_for_view=M8h_FPcAAAAJ:u5HHmVD_u08C <https://onlinelibrary.wiley.com/doi/abs/10.1111/jtsa.12806> <https://arxiv.org/abs/1704.02739>]

Next steps

1. literature review regarding follow-up work on Nicholson et al. (2017, 2020) [all]
2. set up a github repo [AJ: done]
3. maybe contact Jacob Bien, David Matteson or Ines Wilms (she'll be in Recklinghausen at the TRR conference in June 2026) [AJ: done]
4. make BigVar R-package (Nicholson et al., 2026) work and try to replicate Nicholson et al. (2017) simulations and real data applications [AJ: done, we implemented BigTime <https://github.com/ineswilms/bigtime> instead of BigVar]
5. check out BigVAR R code (Nicholson et al., 2026) and get an idea of how easy it would be to break it up w.r.t. adaptive weights and fusion penalties (i.e., difference of coefficients) [AJ: working on it, see <https://github.com/RouvenM/STARX>]
6. brainstorm regarding potential data applications, see above [all] [AJ: done - we decided to use citi bike data]
7. Think about how to simulate data for controlled simulation experiments [all]
8. involve student assistant, Ms. Elina Sukhanova (email: elina.sukhanova@tu-dortmund.de)
9. understand BigTime better, note questions and set up meeting with Ines
10. write down some penaltys (best starting point)
11. A classic fusion penalty approach based on time lags and including reasonable time-decay weights would be:

$$\sum_{i=1}^k \sum_{\ell, \tilde{\ell}=1}^p w_{i, \ell, \tilde{\ell}} \|\Phi_{i.}^{(\ell)} - \Phi_{i.}^{(\tilde{\ell})}\|_2 \quad (3)$$

$$w_{i, \ell, \tilde{\ell}} = e^{-\alpha|\ell - \tilde{\ell}|} \mathbf{1}(\tilde{\ell} > \ell), \quad (4)$$

where i here denotes a single spatial coordinate. This is important as Ines' algorithm deals with penalties that cumulate up over locations, and currently can't deal with penalties that cross locations.

Alternatively, a very first, very simple version of this would just penalize neighboring (w.r.t. time-lags) coefficient matrices and would look like this:

$$\sum_{i=1}^k \sum_{\ell=1}^p \|\Phi_{i.}^{(\ell)} - \Phi_{i.}^{(\ell-1)}\|_2$$

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